

Experiment 7: Analog Beamforming Analysis Using MATLAB

Objectives, MATLAB Programs and Output Images

Objective 1

To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.

MATLAB Program

```
clc;
clear;
close all;

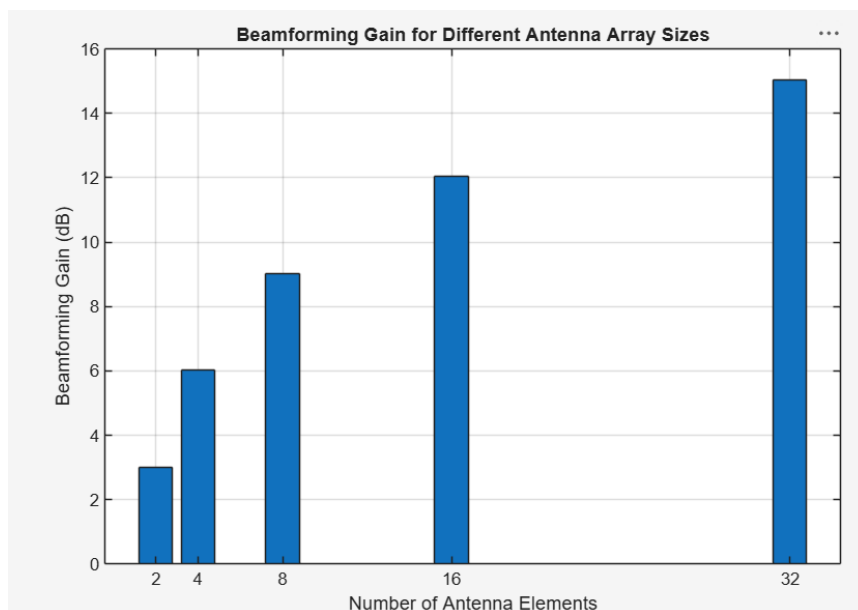
antennas = [2 4 8 16 32];           % Number of Antenna Elements
gain = 10*log10(antennas);           % Beamforming Gain (dB)

figure
bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```

Output Image – Objective 1



Objective 2

To compare the beam width of an antenna array for different beamforming configurations using MATLAB.

MATLAB Program

```
clc;
clear;
close all;

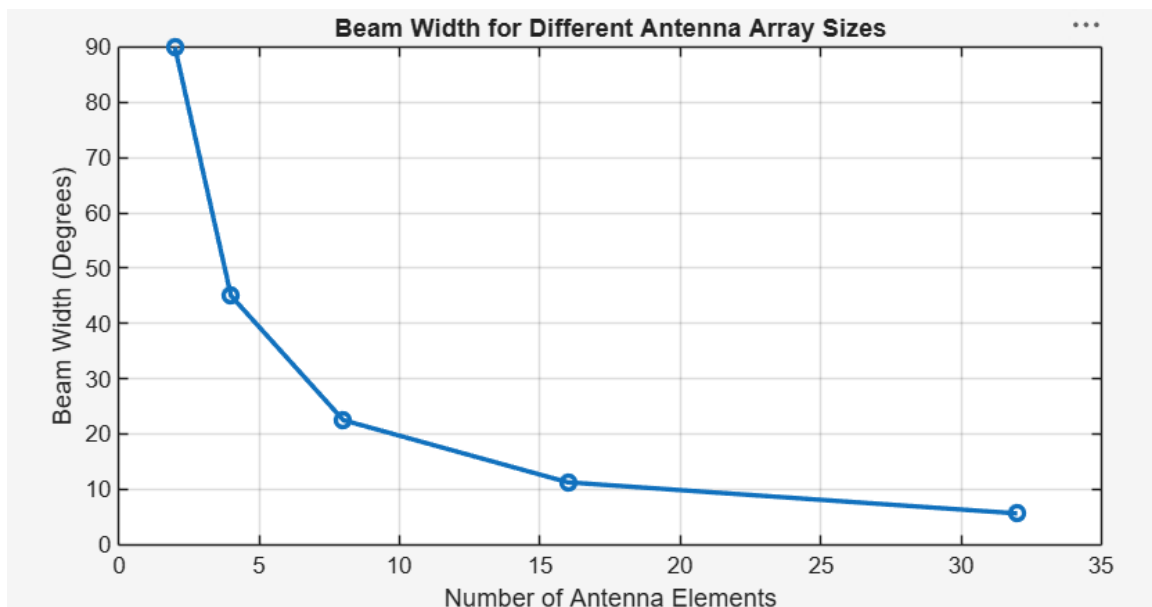
antennas = [2 4 8 16 32]; % Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure
plot(antennas, beamwidth, 'o-', 'LineWidth', 2)

grid on

xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')
```

Output Image – Objective 2



Objective 3

To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

MATLAB Program

```
clc;
clear;
close all;

theta = [-60 -30 0 30 60];      % Main Lobe Direction (degrees)
gain = [1 1 1 1 1];            % Constant normalized gain

figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')
```

Output Image – Objective 3

